

1. Create a Schema using SQL code. Import data to database

Navigator

SCHEMAS

Filter objects

- librarymanagement
- mydb
 - pandemic**
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - sys

SQL File 3*

```
1 /*1. Create a schema*/
2 create schema pandemic;
3 use pandemic;
```

Limit to 1000 rows

Output

#	Time	Action	Message
1	19:27:21	create schema pandemic	1 row(s) affected

Administration Schemas

Information

Schema: **pandemic**

Object Info Session

Navigator

SCHEMAS

Filter objects

- librarymanagement
- mydb
 - pandemic**
 - infectious_cases
 - Columns
 - Entity
 - Code
 - Year
 - Number_yaws
 - polio_cases
 - cases_guinea_worm
 - Number_rabies
 - Number_malaria
 - Number_hiv
 - Number_tuberculosis
 - Number_smallpox
 - Number_cholera_cases
 - Indexes
 - Foreign Keys
 - Triggers
 - Views
 - Stored Procedures
 - Functions
 - sys

SQL File 3*

```
1 /*1. Create a schema*/
2 create schema pandemic;
3 use pandemic;
```

Limit to 1000 rows

Output

#	Time	Action	Message
3	19:29:28	SHOW DATABASES	OK
4	19:29:29	SHOW SESSION VARIABLES LIKE 'lower_case_table_names'	OK
5	19:29:29	SHOW TABLES FROM 'pandemic' like 'infectious_cases'	OK
6	19:29:31	CREATE TABLE 'pandemic'.infectious_cases ('Entity' text, 'Code' text, 'Year' int, 'Number_yaws' text, 'polio_case...	OK
7	19:29:31	PREPARE stmt FROM 'INSERT INTO 'pandemic'.infectious_cases ('Entity','Code','Year','Number_yaws','polio_c...	OK
8	19:30:11	DEALLOCATE PREPARE stmt	OK
9	19:31:22	SELECT Entity FROM 'pandemic'.infectious_cases' LIMIT 0, 1000	1000 row(s) returned
10	19:31:26	SELECT polio_cases FROM 'pandemic'.infectious_cases' LIMIT 0, 1000	1000 row(s) returned
11	19:31:28	SELECT Number_malaria FROM 'pandemic'.infectious_cases' LIMIT 0, 1000	1000 row(s) returned

Administration Schemas

Information

Column: **Number_malaria**

Collation: utf8mb4_0900_ai_ci

Definition:

Number_malaria text

Object Info Session

2. Normalize table to 3NF

Navigator

SCHEMAS

- librarymanagement
- mydb
- pandemic
 - Tables
 - countries

Table: countries

Columns:

- id int AI PK
- Entity varchar(255)
- Code varchar(10)

Object Info Session

SQL File 1

```
3 /*
4  * create schema pandemic;
5  * use pandemic;
6  */
7
8 2. Normalize table to 3NF
9 -----
10 /*
11 -- create a table for countries
12 * create table countries(
13     id int auto_increment primary key,
14     Entity varchar(255),
15     Code varchar(10)
16 );
17
18 * insert into countries(Entity, Code)
19 select distinct Entity, Code
20 from infectious_cases;
21
22 -- create table of sickness
23 * create table infectious_cases_norm(
24     id int auto_increment primary key,
25     country_id int,
26     year int,
27     Number_yaws double,
28     polio_cases double,
29     cases_guinea_worm double,
30     Number_rabies double,
31     Number_malaria double,
32     Number_hiv double,
33     Number_tuberculosis double,
34     Number_smallpox double,
35     Number_cholera_cases double,
36     foreign key (country_id) references countries(id)
37 );
38
```

Output

#	Time	Action	Message
7	19:29:31	PREPARE stmt FROM 'INSERT INTO `pandemic`.`infectious_cases` (`Entity`,`Code`,`Year`,`Number_yaws`,`polio_c...	OK
8	19:30:11	DEALLOCATE PREPARE stmt	OK
9	19:31:22	SELECT Entity FROM `pandemic`.`infectious_cases` LIMIT 0, 1000	1000 row(s) returned
10	19:31:26	SELECT polio_cases FROM `pandemic`.`infectious_cases` LIMIT 0, 1000	1000 row(s) returned
11	19:31:28	SELECT Number_malaria FROM `pandemic`.`infectious_cases` LIMIT 0, 1000	1000 row(s) returned
12	19:53:32	create table countries(id int auto_increment primary key, Entity varchar(255), Code varchar(10))	0 row(s) affected
13	19:54:43	insert into countries(Entity, Code) select distinct Entity, Code from infectious_cases	195 row(s) affected Records: 195 Duplicates: 0 Warnings: 0
14	19:55:25	SELECT * FROM pandemic.countries LIMIT 0, 1000	195 row(s) returned
15	20:00:24	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	0 row(s) affected

Navigator

SCHEMAS

- librarymanagement
- mydb
- pandemic
 - Tables
 - countries
 - infectious_cases
 - infectious_cases_norm

No object selected

SQL File 3

```
22 -- create table of sickness
23 * create table infectious_cases_norm(
24     id int auto_increment primary key,
25     country_id int,
26     year int,
27     Number_yaws double,
28     polio_cases double,
29     cases_guinea_worm double,
30     Number_rabies double,
31     Number_malaria double,
32     Number_hiv double,
33     Number_tuberculosis double,
34     Number_smallpox double,
35     Number_cholera_cases double,
36     foreign key (country_id) references countries(id)
37 );
38
39 -- insert into table
40 * insert into infectious_cases_norm(
41     country_id, year, Number_yaws, polio_cases, cases_guinea_worm,
42     Number_rabies, Number_malaria, Number_hiv, Number_tuberculosis,
43     Number_smallpox, Number_cholera_cases
44 )
45 select
46     c.id, t.year,
47     nullif(t.Number_yaws, '') as Number_yaws,
48     nullif(t.polio_cases, '') as polio_cases,
49     nullif(t.cases_guinea_worm, '') as cases_guinea_worm,
50     nullif(t.Number_rabies, '') as Number_rabies,
51     nullif(t.Number_malaria, '') as Number_malaria,
52     nullif(t.Number_hiv, '') as Number_hiv,
53     nullif(t.Number_tuberculosis, '') as Number_tuberculosis,
54     nullif(t.Number_smallpox, '') as Number_smallpox,
55     nullif(t.Number_cholera_cases, '') as Number_cholera_cases
56 from infectious_cases t
57 join countries c on t.Entity = c.Entity
```

Output

#	Time	Action	Message
20	20:08:23	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	0 row(s) affected
21	20:08:44	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1136. Column count doesn't match value count at row 1
22	20:09:16	SELECT * FROM pandemic.infectious_cases_norm LIMIT 0, 1000	0 row(s) returned
23	20:10:39	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
24	20:11:15	DROP TABLE `pandemic`.`infectious_cases_norm`	0 row(s) affected
25	20:11:20	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	0 row(s) affected
26	20:11:46	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	Error Code: 1050. Table 'infectious_cases_norm' already exists
27	20:12:00	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
28	20:17:55	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	7271 row(s) affected Records: 7271 Duplicates: 0 Warnings: 0

Navigator

SCHEMAS

Filter objects

librarymanagement

mysd

pandemic

Tables

countries

infectious_cases

infectious_cases_norm

Views

Stored Procedures

Functions

sys

SQL File 3*

Limit to 1000 rows

```

35 Number_cholera_cases double,
36 foreign key (country_id) references countries(id)
37 );
38
39 -- insert into table
40 insert into infectious_cases_norm(
41 select
42 c.id, t.year,
43 nullif(t.Number_yaws, '') as Number_yaws,
44 nullif(t.polio_cases, '') as polio_cases,
45 nullif(t.cases_guinea_worm, '') as cases_guinea_worm,
46 nullif(t.Number_rabies, '') as Number_rabies,
47 nullif(t.Number_malaria, '') as Number_malaria,
48 nullif(t.Number_hiv, '') as Number_hiv,
49 nullif(t.Number_tuberculosis, '') as Number_tuberculosis,
50 nullif(t.Number_smallpox, '') as Number_smallpox,
51 nullif(t.Number_cholera_cases, '') as Number_cholera_cases
52 from infectious_cases t
53 join countries c on t.Entity = c.Entity;
54
55 -- check count of rows
56 select count(*) from infectious_cases;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

count(*)

7271

Administration Schemas

Information

No object selected

Object Info Session

Action Output

#	Time	Action	Message
21	20:08:44	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1136. Column count doesn't match value count at row 1
22	20:09:16	SELECT * FROM pandemic.infectious_cases_norm LIMIT 0, 1000	0 row(s) returned
23	20:10:39	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
24	20:11:15	DROP TABLE 'pandemic'.infectious_cases_norm	0 row(s) affected
25	20:11:20	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	0 row(s) affected
26	20:11:45	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws...	Error Code: 1050. Table 'infectious_cases_norm' already exists
27	20:12:00	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
28	20:17:55	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_ra...	7271 row(s) affected Records: 7271 Duplicates: 0 Warnings: 0
29	20:19:39	select count(*) from infectious_cases LIMIT 0, 1000	1 row(s) returned

3. Analyze data

MySQL Workbench

Local instance MySQL80

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Navigator

SCHEMAS

Filter objects

librarymanagement

mysd

pandemic

Tables

countries

infectious_cases

infectious_cases_norm

Views

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Functions

sys

SQL File 3*

Limit to 1000 rows

```

57 join countries c on t.Entity = c.Entity;
58
59 -- check count of rows
60 select count(*) from infectious_cases;
61
62 /*
63 3. Analyze data
64 */
65
66 select
67 c.Entity,
68 c.Code,
69 avg(n.Number_rabies) as avg_rabies,
70 min(n.Number_rabies) as min_rabies,
71 max(n.Number_rabies) as max_rabies,
72 sum(n.Number_rabies) as sum_rabies
73 from infectious_cases_norm n
74 join countries c on n.country_id = c.id
75 where n.Number_rabies is not null and n.Number_rabies != ''
76 group by c.Entity, c.Code
77 order by avg_rabies desc
78 limit 10;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

#	Entity	Code	avg_rabies	min_rabies	max_rabies	sum_rabies
1	World	OWID_WRL	20192.370366666666	14075.508	24744.658	605771.111
2	India	IND	8996.172379999999	5428.8726	11121.142	257975.18839999999
3	Pakistan	PAK	1582.1696266666663	1177.1449	1881.7257	47465.08879999999
4	Nigeria	NGA	1335.4154866666667	1073.9458	1441.1188	40062.46460000001
5	China	CHN	1247.606753	744.251	2075.593	37437.20199
6	Ethiopia	ETH	1145.1725233333333	758.0497	1323.4026	34355.17567
7	Myanmar	MMR	972.9036866666668	560.49684	1449.8279	29187.150600000004
8	Bangladesh	BGD	785.7088230000002	180.40259	1734.7075	23571.264690000004
9	Nepal	NPL	661.2508643333334	338.1352	1041.6312	19837.529300000003
10	Philippines	PHL	618.6910056666668	358.0602	898.071	18560.730170000003

Result 2 x

Output

Action Output

#	Time	Action	Message
22	20:09:16	SELECT * FROM pandemic.infectious_cases_norm LIMIT 0, 1000	0 row(s) returned
23	20:10:39	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_rabies, Num...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
24	20:11:15	DROP TABLE 'pandemic'.infectious_cases_norm	0 row(s) affected
25	20:11:20	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws double, ...	0 row(s) affected
26	20:11:45	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws double, ...	Error Code: 1050. Table 'infectious_cases_norm' already exists
27	20:12:00	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_rabies, Num...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
28	20:17:55	insert into infectious_cases_norm(country_id, year, Number_yaws, polio_cases, cases_guinea_worm, Number_rabies, Num...	7271 row(s) affected Records: 7271 Duplicates: 0 Warnings: 0
29	20:19:39	select count(*) from infectious_cases LIMIT 0, 1000	1 row(s) returned
30	20:24:55	select c.Entity, c.Code, avg(n.Number_rabies) as avg_rabies, min(n.Number_rabies) as min_rabies, max(n.Number_r...	10 row(s) returned

Administration Schemas

Information

No object selected

Object Info Session

4. Date difference

MySQL Workbench

Local instance MySQL80

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Tables

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sys

SQL File 3

3. Analyze data

```

63 3. Analyze data
64 =====
65 */
66 • select
67     c.Entity,
68     c.Code,
69     avg(n.Number_rabies) as avg_rabies,
70     min(n.Number_rabies) as min_rabies,
71     max(n.Number_rabies) as max_rabies,
72     sum(n.Number_rabies) as sum_rabies
73 from infectious_cases_norm n
74 join countries c on n.country_id = c.id
75 where n.Number_rabies is not null and n.Number_rabies != ''
76 group by c.Entity, c.Code
77 order by avg_rabies desc
78 limit 10;
79
80 /*
81 4. Years difference
82 =====
83 */
84 • select

```

Result Grid

Filter Rows:

Exports

Wrap Cell Contents

Fetch rows:

	year	first_jan_date	current_date_val	year_diff
▶	1980	1980-01-01	2026-02-02	46
	1981	1981-01-01	2026-02-02	45
	1982	1982-01-01	2026-02-02	44
	1983	1983-01-01	2026-02-02	43
	1984	1984-01-01	2026-02-02	42
	1985	1985-01-01	2026-02-02	41
	1986	1986-01-01	2026-02-02	40
	1987	1987-01-01	2026-02-02	39
	1988	1988-01-01	2026-02-02	38
	1989	1989-01-01	2026-02-02	37

Result 3

Output

Action Output

#	Time	Action	Message
23	20:10:39	insert into infectious_cases_norm(country_id, year, Number_yaws, polo_cases, cases_guinea_wom, Number_rabies, Num...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
24	20:11:15	DROP TABLE `pandemic`.`infectious_cases_norm`	0 row(s) affected
25	20:11:20	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws double, ...	0 row(s) affected
26	20:11:46	create table infectious_cases_norm(id int auto_increment primary key, country_id int, year int, Number_yaws double, ...	Error Code: 1050. Table 'infectious_cases_norm' already exists
27	20:12:00	insert into infectious_cases_norm(country_id, year, Number_yaws, polo_cases, cases_guinea_wom, Number_rabies, Num...	Error Code: 1265. Data truncated for column 'Number_yaws' at row 1
28	20:17:55	insert into infectious_cases_norm(country_id, year, Number_yaws, polo_cases, cases_guinea_wom, Number_rabies, Num...	7271 row(s) affected Records: 7271 Duplicates: 0 Warnings: 0
29	20:19:39	select count(*) from infectious_cases LIMIT 0, 1000	1 row(s) returned
30	20:24:55	select c.Entity, c.Code, avg(n.Number_rabies) as avg_rabies, min(n.Number_rabies) as min_rabies, max(n.Number_r...	10 row(s) returned
31	20:29:29	select year, makedate(year, 1) as first_jan_date, curdate() as current_date_val, timestampdiff(year, makedate(year, 1)...	10 row(s) returned

Administration

Schemas

Information

No object selected

Object Info

Session

5. Own function

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- librarymanagement
- mydb
 - Tables
 - countries
 - infectious_cases
 - infectious_cases_norm
 - Views
 - Stored Procedures
 - Functions
- sys

SQL File 3

```

89 from infectious_cases_norm
90 limit 10;
91
92 /*
93 5. Own function
94 =====
95 */
96 • drop function if exists CalculateYearDiff;
97
98 DELIMITER //
99
100 • create function CalculateYearDiff(input_year int)
101 returns int
102 deterministic
103 begin
104 declare result int;
105 set result = timestampdiff(year, makedate(input_year, 1), curdate());
106 return result;
107 end //
108
109 DELIMITER ;
110
111 • select
112 year,
113 CalculateYearDiff(year) as year_diff_func
114 from infectious_cases_norm
115 limit 10;

```

Result Grid

year	year_diff_func
1980	46
1981	45
1982	44
1983	43
1984	42
1985	41
1986	40
1987	39
1988	38
1989	37

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message
45	20:48:10	create function CalculateYearDiff(input_year int) returns int deterministic begin declare result int	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL
46	20:48:34	drop function if exists CalculateYearDiff	0 row(s) affected, 1 warning(s): 1305 FUNCTION pandemic.CalculateYearDiff does not exist
47	20:49:36	drop function if exists CalculateYearDiff	0 row(s) affected, 1 warning(s): 1305 FUNCTION pandemic.CalculateYearDiff does not exist
48	20:50:32	create function CalculateYearDiff(input_year int) returns int deterministic begin declare result int; set result = timestampdiff(ye...	0 row(s) affected
49	20:50:40	select year, CalculateYearDiff(year) as year_diff_func from infectious_cases_norm limit 10	10 row(s) returned

Object Info Session